

JUNIOR DATA SCIENTIST | RESEARCH ANALYST (DATA SCIENCE & ECONOMICS)

Isabella Cappiello | Nationality: Italian

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PROFESSIONAL PROFILE

Data Science Double Degree Master's Candidate (IT/FR – Unistra) with a GPA of **29.30/30** and a **110/110 Cum Laude** Bachelor's in Economics. Expert in engineering **Python/R** pipelines to harmonize large-scale institutional datasets (ISTAT/MEF) for **Spatio-Temporal ML** and **policy evaluation**. Specialized in **Causal Inference (Staggered DiD)** and delivering technical **knowledge transfer** to a stakeholder base of 200+ users in multicultural environments.

PROJECTS & RESEARCH

- Spatio-Temporal Analysis: Impact of "1-Euro Houses"** | **R, Shiny** | *Jan 2026 – Present, Brescia, Italy*
- Engineered an R pipeline to harmonize 20+ years of ISTAT/MEF data (7,900+ municipalities), resolving administrative mergers via recursive spatial algorithms.
 - Quantified socio-economic impacts using Staggered Difference-in-Differences (DiD) and causal inference.
 - Developed a Shiny Dashboard (sf, leaflet) for real-time geospatial visualization, performing rigorous longitudinal testing to identify effects on regional vitality.
- Predictive Analytics: Student Mental Health Classifier** | **Python** | *Dic 2024 – Jan 2025, Strasbourg, France*
- Developed an NLP/ML framework using Regex and Random Forest to classify depression markers.
 - Optimized precision-recall via SMOTE and cost-sensitive learning to mitigate class imbalance.
 - Executed hyperparameter tuning to maximize model generalizability and diagnostic sensitivity.
- Quantitative Finance: ESG & Risk-Adjusted Performance** | **R** | *July 2023 – Oct 2023, Brescia, Italy*
- Quantified risk-adjusted alpha using CAPM and HAC Standard Errors to model volatility trajectories.
 - Validated market trends via intensive hypothesis testing (t-tests, F-tests) and ESG risk assessment.
 - Engineered an R pipeline ensuring robustness against heteroskedasticity in longitudinal data.

WORK EXPERIENCE

- STEM Educator: Digital Literacy & Data Analytics** | *Public Schools | May 2025 – Present*
- Upskilled 200+ students in Advanced Spreadsheet Modeling, Pivot Tables, and computational logic.
 - Standardized interdisciplinary workflows focusing on Data Integrity and technical reasoning.
 - Led IT knowledge transfer via Google Workspace, ensuring 100% digital literacy.

EDUCATION

- Double Master's: Analytics & Data Science** | *Univ. of Brescia (IT) & Strasbourg (FR) | Sept 2023 – Present*
- GPA: 29.30/30 | International Mobility: Strasbourg (Sept 2024 – March 2025).
 - Core Competencies: Advanced Econometrics, ML, DL, RL, Spatio-Temporal Statistics.
 - Master's Thesis: Impact of "1-Euro Houses" (Focus: Staggered DiD & Geospatial Analysis in R).
- Bachelor's in Economics & Digital Business** | *University of Brescia (IT) | Sept 2020 – Oct 2023*
- Result: 110/110 Cum Laude | Quantitative Focus: Empirical Economics, OR, Math Stats.
 - Thesis: ESG vs. Traditional Investments (CAPM & Econometric Assessment in R).

TECHNICAL STACK & SKILLS

- Programming:** R (tidyverse, sf, leaflet), Python (scikit-learn, pandas), SQL, Git/GitHub, R Shiny, Jupyter.
- Methods:** Causal Inference, Staggered DiD, Econometrics, ML/DL/RL, Geospatial Analysis, Reproducible Research.
- Certification:** Python Accelerator (Advanced Workflows & Modularity) – Expected May 2026.
- Professional:** Knowledge Transfer, Data-Driven Strategy, Ethics & Governance.
- Languages:** English (C1), Italian (Native), French (A2).

LEADERSHIP & COMMUNITY ENGAGEMENT

- Executive Board (Finance & Strategy)** | *Elia Marini Band*: Elected Treasurer managing institutional budgeting and Social Media Manager for a 30-member ensemble; active Clarinettist.
- Advocacy & Performance** | *Women Wind Orchestra*: Supporting member for cancer research and gender equality initiatives; Clarinettist in 60-member symphonic ensembles.